Lab 9: Graph, PowerShell, remote actions and Endpoint Analytics

Duration: 115 minutes
 Alignment: LO3 | LO4 | K3 | K5 | A5 | A7
 Deliverable: Least-privilege automation transcript, paginated export, remote-action evidence and KPI readout

Safety and prerequisites

- Use only the training tenant, test users and test devices supplied by the trainer.
- Confirm the tenant ID before any change.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate, reprovision or other destructive actions.

Authenticate with least privilege

1. Open PowerShell 7 and connect with Connect-MgGraph using only the lab read scopes.
2. Run Get-MgContext and verify TenantId, Account and Scopes.
3. Record the Graph v1.0 endpoint and selected resource.
4. Do not request write permissions until the trainer authorizes the change task.

Acceptance check: capture the configured state, observed result, timestamp/correlation ID and interpretation.

Export all managed devices

1. Request the managedDevices collection with selected fields.
2. Follow every @odata.nextLink until no next page remains.
3. Export device ID, name, OS, compliance and last-sync values to CSV.
4. Compare the exported count with the service report.

Acceptance check: capture the configured state, observed result, timestamp/correlation ID and interpretation.

Perform a controlled action

1. Resolve one lab device by stable device ID.
2. Preview the intended action and confirm ownership.
3. Run the approved sync or rotate-key action.
4. Capture the audit actor, time and resulting status.

Acceptance check: capture the configured state, observed result, timestamp/correlation ID and interpretation.

Interpret analytics

1. Open Endpoint Analytics and record the current startup, application reliability and restart signals.
2. Segment by the trainer-provided device model or OS cohort.
3. Identify one outlier and propose a bounded proactive remediation.
4. Define the before/after metric that will prove improvement.

Acceptance check: capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under evidence/ and complete evidence/README.md. Do not include passwords, recovery keys, tokens, tenant secrets or unrelated personal data.

Source trace

Aligned to Microsoft MD-102 skills measured 24 July 2026 and adapted from MicrosoftLearning repository commit 25eaad2a79548c4767ea45eae8d08649622c583a.